

BiFocus: Evidence-Grounded Patch and Query-Token Calibration for Visual Document Retrieval

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Abstract

Visual document retrieval has recently shifted from OCR-heavy document parsing pipelines toward screenshot-based late-interaction retrievers such as ColPali. These models preserve fine-grained token-patch matching, but their supervision remains misaligned with their scoring mechanism: relevance is accumulated over query-token MaxSim scores, while training signals are typically page-level and, at best, patch-side. We argue that this creates a supervision asymmetry: existing methods can teach visual retrievers *where* to look on the page, but provide limited explicit supervision for *which query tokens* should drive the match. We propose BiFocus, an evidence-grounded late-interaction framework that repurposes explicit evidence boxes from BBox-DocVQA to calibrate both page-side patch grounding and query-side token contribution. The framework first converts bounding-box evidence into soft patch-level targets, then induces token-level evidence credits from token-patch alignment and uses them to train a lightweight query-token scorer. At inference time, no evidence boxes or teacher models are required; the model computes a calibrated weighted late-interaction score that emphasizes query tokens supported by document evidence while preserving the original page-level retrieval interface. On ViDoRe V2, BiFocus improves ColPali-v1.3 by +6.15 nDCG@1 and +3.80 nDCG@5, and improves ColQwen2.5-v0.2 by +7.12 nDCG@1 and +4.72 nDCG@5 under the same-backbone setting. On the more extractive and saturated ViDoRe V1 benchmark, BiFocus preserves the strong baseline performance. Code is available at <https://github.com/YUEcjy13/BiFocus>.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; *Evaluation of retrieval results.*

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Keywords

visual document retrieval, late interaction, query-token weighting, evidence grounding, visually rich documents

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1 Introduction

Visual document retrieval (VDR) aims to retrieve document pages that satisfy a textual information need. Unlike plain-text retrieval, visually rich pages distribute meaning across words, layout, tables, figures, charts, formulas, and spatial composition. OCR-heavy pipelines must parse and reconstruct these structures before retrieval, and each step can discard information or introduce errors. Recent screenshot-based retrievers therefore encode rendered page images directly and match them to text queries in a shared representation space [10, 22]. Among them, ColPali is a strong backbone because it combines a vision-language model with ColBERT-style late interaction, allowing each query token to match fine-grained page patches rather than a single compressed page vector [10, 17].

Late interaction is powerful, but its supervision is often mismatched with its scoring rule. A ColPali-style model represents a query as token embeddings and a page as patch embeddings, scores each query token by its maximum similarity to any page patch, and sums these token-wise scores. In other words, the model is trained with page-level labels, may be guided by patch-level signals, but ultimately scores pages through query-token decisions. Training, however, is usually page-level: the model is told whether a page is relevant, not which visual content supports the match. Recent work such as AGREE addresses this limitation by introducing patch-level local supervision from multimodal large language model (MLLM) attention, showing that query-conditioned region guidance improves VDR on the challenging ViDoRe V2 benchmark [8, 24].

As illustrated in Fig. 1, patch grounding alone does not fully match the structure of late interaction. Existing fine-grained VDR methods mainly correct the *page side*: they teach the retriever where

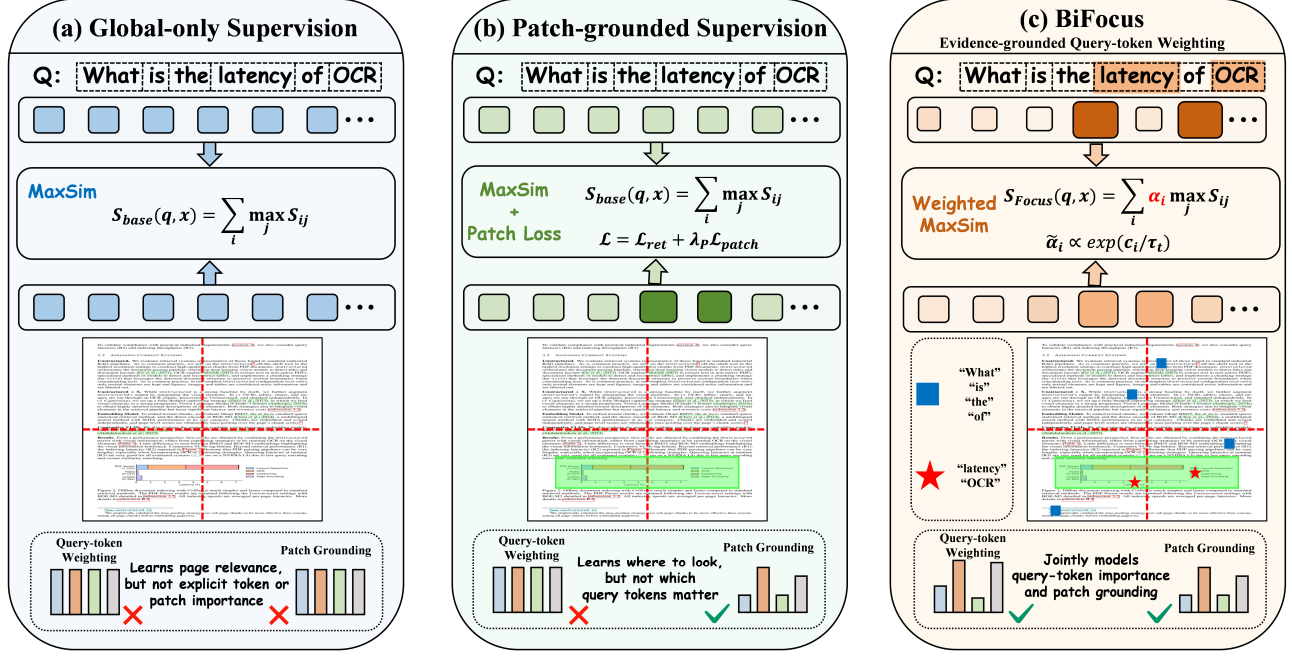


Figure 1: Motivation of BiFocus. Global supervision optimizes page-level relevance but leaves patch evidence and query-token contribution implicit. Patch-grounded supervision teaches the retriever *where to look*, but the final MaxSim score is still aggregated over query tokens with little explicit control. BiFocus uses explicit evidence boxes to calibrate both page-side grounding and query-side token contribution, yielding evidence-aware weighted late interaction.

relevant evidence may lie. Yet the final relevance score is accumulated over the *query side*. Every query token independently searches for its best page patch, and treating all query tokens as equally reliable is often unreasonable. Domain terms, entities, numerical attributes, and relation words usually define the evidence, whereas function words and generic context tokens contribute less to the retrieval decision. If training only constrains page-level relevance or patch-side saliency, the retriever may learn *where to look* without learning *which query tokens should drive the match*. We refer to this mismatch as a **supervision asymmetry** in late-interaction VDR.

This asymmetry suggests that grounded visual retrieval should be two-sided. Query-token importance has been studied in text-based multi-vector retrieval [2, 16, 31, 32], but these methods do not answer how token importance should be learned when relevance depends on spatially localized evidence inside a rendered page. In VDR, a token can be globally rare yet irrelevant to the supporting evidence of a particular page, or visually decisive despite being short, numeric, or relational. Token weighting should therefore be tied to whether a token actually aligns with the evidence regions that support the query–page match.

A second question is where local supervision should come from. AGREE demonstrates the value of patch-level guidance, but its labels are proxy signals derived from MLLM attention maps [8]. In contrast, BBox-DocVQA provides explicit bounding-box evidence for document visual question answering, where each question–answer instance is grounded in coherent semantic regions such as paragraphs, tables, and figures [44]. Although BBox-DocVQA was

not designed for retrieval, its annotations provide the page-internal evidence that late-interaction retrievers lack. This makes the supervision practically useful: boxes expose localized support during training, while the deployed retriever still operates on ordinary query–page pairs without OCR regions, answer annotations, or teacher attention.

We propose **BiFocus**, an evidence-grounded late-interaction framework for VDR. First, we project BBox-DocVQA evidence boxes onto the ColPali patch grid and construct soft patch-level evidence targets, teaching the retriever to concentrate query-conditioned similarity on supporting page regions. Second, we reuse the same evidence target to derive query-token credits from token-specific patch alignment: a token receives higher credit when its patch distribution overlaps with grounded evidence. These credits supervise a lightweight query-token scorer. At inference time, the scorer predicts token weights from the query representation alone, and the final relevance score is computed by calibrated weighted late interaction. No bounding boxes, OCR regions, or teacher MLLM attention maps are required at inference.

Conceptually, BiFocus turns fine-grained VDR supervision from page-only grounding into joint token–patch calibration: the same evidence boxes indicate where support lies on the page and, through token–patch overlap, which query tokens should contribute more to the final score. In implementation, we use a stable staged optimization schedule: the retriever is first adapted with bbox-derived patch grounding, and the query-token scorer is then calibrated from evidence-induced token credits on top of the grounded retriever.

This schedule follows the causal structure of the supervision, since reliable token credits are best induced after token–patch maps have learned to align with explicit evidence.

We evaluate BiFocus on ViDoRe V2 and ViDoRe V1. ViDoRe V2 is our primary benchmark because it reduces the extractive-query bias and saturation of the original ViDoRe benchmark and emphasizes more realistic, non-extractive, long-form, and cross-document retrieval scenarios [24]. BiFocus substantially improves both ColPali-v1.3 and ColQwen2.5-v0.2 on ViDoRe V2 under same-backbone adaptation, while remaining competitive on the more extractive and saturated ViDoRe V1 benchmark.

Our contributions are fourfold:

- We identify a supervision asymmetry in late-interaction VDR: existing methods can ground page patches, while query-token contribution often remains implicit despite being central to MaxSim aggregation.
- We repurpose explicit BBox-DocVQA evidence boxes for retrieval by converting them into soft patch targets aligned with the screenshot patch grid, without requiring boxes, OCR regions, or teacher attention at inference time.
- We introduce BiFocus, a unified evidence-grounded calibration framework that combines bbox-to-patch grounding with evidence-induced query-token weighting for screenshot-based late-interaction retrieval.
- We conduct comprehensive experiments on ViDoRe V2 and V1 with ColPali and ColQwen backbones, showing strong V2 gains, stable V1 behavior, informative ablations, and a lightweight inference pipeline with unchanged page-index storage.

2 Related Work

Screenshot-based visual document retrieval. Document retrieval has long relied on text-centric pipelines, from lexical matching to dense encoders and neural rerankers [4, 9, 23, 30, 34, 35, 39]. For visually rich documents, however, OCR, PDF parsing, layout detection, and table reconstruction can discard layout and visual evidence before retrieval. Screenshot-based VDR instead treats the rendered page image as the retrieval unit [10, 22]. ColPali is a representative late-interaction retriever: it adapts a vision–language model to produce page-patch multi-vectors and matches them with query-token vectors using ColBERT-style MaxSim scoring [10, 17]. ViDoRe established page-level visual retrieval, while ViDoRe V2 stresses less extractive, more realistic, and often cross-document information needs [10, 24]. Our work keeps the screenshot page interface but studies how evidence supervision should be injected into a model whose final score is accumulated over query-token matches.

Fine-grained supervision and grounded document evidence. Page-level relevance labels are often too coarse for visually complex retrieval, motivating region-level retrieval, rationale propagation, and local similarity supervision [13, 21, 37, 43]. The closest work is AGREE, which uses MLLM attention as proxy local supervision and shows that query-conditioned region guidance improves ViDoRe V2 retrieval [8]. This confirms the value of grounding, but mainly corrects the *page side*. We instead use explicit evidence boxes and further transform grounded token–patch alignment into query-token contribution learning. Our supervision comes from

BBox-DocVQA, which associates document questions with evidence boxes over coherent regions such as paragraphs, tables, and figures [44]. Unlike document understanding resources designed for extraction, classification, or QA [1, 6, 14, 18, 20, 25, 27–29, 38, 40–42], we repurpose boxes as page-internal evidence supervision for training and calibrating a retriever.

Late interaction and query-token contribution. Late interaction was introduced by ColBERT, where each query token retrieves its strongest matching document token and token-wise scores are summed [17]. Later systems improve efficiency, distillation, and first-stage retrieval quality [11, 12, 19, 36]. Recent text-retrieval studies revisit the uniform-token assumption by learning token importance, modeling token relations, or adding token-level attention to multi-vector scoring [2, 16, 31, 32]. These methods motivate token-aware aggregation, but operate primarily in text retrieval. BiFocus brings this idea into screenshot-based VDR by inducing query-token weights from how each token aligns with explicit visual evidence, rather than from hand-crafted lexical priors.

Complementary multi-view representation learning preserves reliable evidence under heterogeneous or incomplete views through structure-aware generation, shared-space aggregation, diffusion-contrastive recovery, and global–local sequence modeling [46–49].

Multimodal grounding. Screenshot-based retrieval benefits from general-purpose vision–language pre-training such as CLIP and ALIGN [15, 33], SigLIP [45], PaLI and PaliGemma [5, 7], and Qwen-VL and InternVL [3, 50]. BiFocus is aligned with grounded multimodal learning, but preserves the page-level late-interaction retrieval interface: evidence boxes are used only during training or calibration, and no generated descriptions, teacher attention maps, or region-level retrieval are required at inference time.

3 Method

We present BiFocus, an evidence-grounded late-interaction framework for visual document retrieval. A ColPali-style retriever computes relevance by matching each query token to page patches and aggregating token-wise MaxSim scores, yet standard supervision is either page-level relevance or, at most, patch-side saliency. BiFocus follows three design principles: local evidence should shape the internal patch map rather than replace page-level ranking, token contribution should be induced from grounded token–patch alignment rather than lexical salience alone, and evidence annotations should be used only during training or calibration so that deployment keeps the original retrieval interface.

Figure 2 gives an overview. Given a query and a page screenshot, the retriever produces query-token embeddings and page-patch embeddings. Evidence boxes are projected onto the patch grid to form a soft evidence target. This target supervises a query-conditioned patch relevance map and is then reused to derive token-level credits from token-specific patch distributions. A lightweight query-token scorer learns to predict these credits, and inference replaces equal MaxSim aggregation with calibrated weighted late interaction. The supervision graph is unified, while our implementation uses a stable staged schedule: retriever grounding followed by token-scorer calibration.

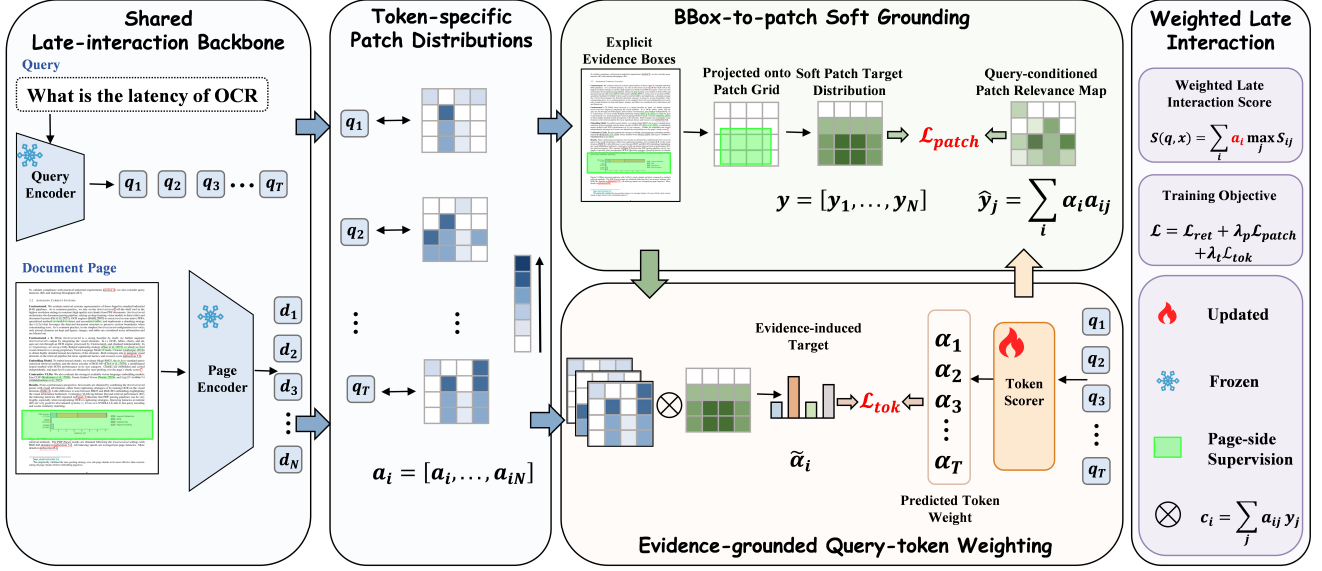


Figure 2: Overview of BiFocus. The method keeps a ColPali-style screenshot retrieval backbone and augments it with two evidence-grounded branches. Explicit evidence boxes are projected onto the page patch grid to supervise a query-conditioned patch relevance map. Token-specific patch distributions are then compared with the evidence target to induce query-token credits, which supervise a lightweight token scorer. At inference time, no boxes are required; the scorer predicts query-token weights and the final page score is computed with calibrated weighted late interaction.

3.1 Late-Interaction Backbone

Let $q = [w_1, \dots, w_T]$ be a query with T non-padding tokens and let x be a document page screenshot. Following ColPali [10], the query encoder maps q into token embeddings

$$Q = [q_1, \dots, q_T] \in \mathbb{R}^{T \times d}, \quad (1)$$

and the page encoder maps x into patch embeddings

$$D = [d_1, \dots, d_N] \in \mathbb{R}^{N \times d}, \quad (2)$$

where N is the number of page patches. With ℓ_2 -normalized embeddings, the token-patch similarity is

$$s_{ij} = \langle q_i, d_j \rangle. \quad (3)$$

The standard late-interaction score is

$$S_{eq}(q, x) = \sum_{i=1}^T \max_{1 \leq j \leq N} s_{ij}. \quad (4)$$

This score preserves fine-grained token-patch matching, but it also implicitly sums query-token contributions with equal strength. BiFocus keeps the same multi-vector backbone and introduces evidence-grounded patch supervision and query-token weighting before computing the final score. When page-level query-page pairs are used, we retain the original contrastive retrieval objective as the global ranking anchor; the local evidence objectives calibrate the internal token-patch behavior.

3.2 BBox-to-Patch Evidence Grounding

The first component teaches the retriever where supporting evidence lies. For a grounded positive query-page pair, let $\mathcal{B} = \{b_k\}_{k=1}^K$ be the BBox-DocVQA evidence boxes [44]. Each box corresponds

to a coherent semantic region, such as a paragraph, table, or figure. Page images are resized to 448×448 and aligned with a 32×32 patch grid, giving $N = 1024$ patches with patch size 14. Let r_j be the spatial support of patch j after resizing. We define the soft evidence target $y = [y_1, \dots, y_N]$ by normalized box-patch overlap:

$$y_j = \frac{1}{Z} \sum_{k=1}^K \frac{|r_j \cap b_k|}{|r_j|}, \quad Z = \sum_{j=1}^N \sum_{k=1}^K \frac{|r_j \cap b_k|}{|r_j|}. \quad (5)$$

Soft overlap is more stable than binary assignment near box boundaries and naturally handles multiple evidence regions. The normalization also makes the local target comparable across pages with different box sizes, so large paragraphs do not dominate the loss merely because they cover more patches. Samples with no positive patch mass are excluded from grounded patch supervision.

To compare the retriever's behavior with y , we construct token-specific patch distributions from the similarity matrix:

$$a_{ij} = \frac{\exp(s_{ij}/\tau_p)}{\sum_{j'=1}^N \exp(s_{ij'}/\tau_p)}, \quad (6)$$

where $\mathbf{a}_i = [a_{i1}, \dots, a_{iN}]$ describes where query token i places its matching mass. These distributions are aggregated into a query-conditioned patch map,

$$\hat{y}_j = \sum_{i=1}^T \beta_i a_{ij}, \quad \sum_{i=1}^T \beta_i = 1. \quad (7)$$

During the initial grounding stage, β_i is uniform, avoiding circular dependence on a token scorer before stable evidence-aligned maps are available.

The theoretical framework only requires a local alignment loss between $\hat{y}$ and y . A strict distributional objective is

$$\mathcal{L}_{\text{patch}}^{\text{KL}} = \text{KL}(y \parallel \hat{y}), \quad (8)$$

but full-distribution matching can over-penalize low-mass background patches when boxes are imperfect. In our implemented patch-grounding stage, we therefore use a cosine-style alignment loss over patch relevance patterns,

$$\mathcal{L}_{\text{patch}} = 1 - \frac{\langle \mathbf{r}, \mathbf{y} \rangle}{\|\mathbf{r}\|_2 \|\mathbf{y}\|_2}, \quad (9)$$

where $\mathbf{r}$ is the retriever-produced patch relevance vector derived from the same query-patch similarities as $\hat{y}$. This objective encourages evidence patches to receive larger relevance while being less sensitive to small deviations over background regions. The patch objective is therefore not an additional retrieval label, but a structural constraint on the internal similarity map of positive query-page pairs: the global retrieval loss anchors page-level ranking, while the local loss shapes where the positive match is expressed.

3.3 Evidence-Induced Query-Token Credits

Patch grounding answers where the model should look, but not which query tokens should be trusted to find that evidence. We therefore derive token-level supervision from grounded token-patch alignment. Given token-specific patch distributions $\mathbf{a}_i$ and the evidence target y , the evidence coverage of token i is

$$c_i = \sum_{j=1}^N a_{ij} y_j. \quad (10)$$

A token receives high credit when it places probability mass on grounded evidence and low credit when its mass falls elsewhere. We convert these credits into a soft token target,

$$\tilde{\alpha}_i = \frac{\exp(c_i / \tau_t)}{\sum_{i'=1}^T \exp(c_{i'} / \tau_t)}, \quad (11)$$

where τ_t controls target sharpness. This target is induced from visual evidence rather than from linguistic heuristics.

To use token weights at inference time, we train a lightweight query-side scorer $g(\cdot)$ that predicts a scalar logit for each query-token embedding,

$$u_i = g(\mathbf{q}_i), \quad \alpha_i = \frac{\exp(u_i / \tau_q)}{\sum_{i'=1}^T \exp(u_{i'} / \tau_q)}. \quad (12)$$

The scorer is trained with

$$\mathcal{L}_{\text{tok}} = \text{KL}(\tilde{\alpha} \parallel \alpha), \quad (13)$$

optionally with light regularization to avoid overly sharp weights on noisy samples. Since $g(\cdot)$ consumes only query-token embeddings, it requires no evidence boxes, document images, or teacher models at inference time. This design separates training-time evidence induction from deployment-time scoring: boxes create token-credit targets on grounded positive pairs, but the learned scorer can later be applied to arbitrary retrieval queries before any candidate page is examined. Thus, query calibration does not change page indexing or leak test-time document evidence into scoring.

3.4 Weighted Late Interaction

After learning the query-token scorer, we replace equal aggregation with weighted late interaction. Let

$$m_i(x) = \max_{1 \leq j \leq N} s_{ij} \quad (14)$$

be the MaxSim score contributed by token i for page x . The evidence-weighted score is

$$S_w(q, x) = \sum_{i=1}^T \alpha_i m_i(x). \quad (15)$$

Because $\sum_i \alpha_i = 1$, S_w is a weighted average over token scores, while S_{eq} in Eq. (4) is a sum over T query tokens. We therefore use a calibrated interpolation,

$$S_\gamma(q, x) = (1 - \gamma) S_{\text{eq}}(q, x) + \gamma S_w(q, x), \quad (16)$$

where $\gamma \in [0, 1]$ controls the strength of query-token calibration. This formulation preserves the strong baseline score while allowing the learned scorer to refine token contributions. The interpolation is important because the scorer is trained as a calibration module rather than a replacement for the late-interaction backbone: $\gamma = 0$ recovers the original retriever, while larger γ values progressively inject evidence-aware token weighting without changing candidate-page embeddings.

3.5 Unified Objective, Optimization, and Inference

The supervision graph of BiFocus can be summarized by the following evidence-grounded objective,

$$\mathcal{L} = \mathcal{L}_{\text{ret}} + \lambda_p \mathcal{L}_{\text{patch}} + \lambda_t \mathcal{L}_{\text{tok}}, \quad (17)$$

where $\mathcal{L}_{\text{ret}}$ is the standard page-level retrieval loss and λ_p, λ_t balance global ranking, page-side grounding, and query-side token calibration. Eq. (17) should be read as the conceptual decomposition of the supervision signals rather than a claim that all parameters are updated jointly in a single optimization step. All reported results use the following two-stage instantiation. First, starting from a public late-interaction checkpoint, we fine-tune the retriever with page-level retrieval supervision and bbox-derived patch grounding. Second, we keep the grounded retriever fixed when computing evidence-induced token credits and train only the lightweight query-token scorer. This staged strategy avoids bootstrapping token targets from an ungrounded retriever: the token scorer is calibrated after the token-patch maps have already learned to align with explicit evidence. Thus, the staged schedule is an optimization choice for realizing Eq. (17), not a conceptual separation of the two branches.

At inference time, BiFocus uses the same inputs as ColPali: a textual query and candidate page screenshots. No evidence boxes, OCR boxes, MLLM attention maps, or external teachers are required. We encode the query and pages with the late-interaction backbone, predict query-token weights with the learned scorer, and compute the calibrated weighted score in Eq. (16). Thus, the method changes how the retriever is trained and how token contributions are aggregated, but keeps the page-level retrieval interface unchanged.

4 Experiments

We evaluate BiFocus around four questions: whether evidence-grounded adaptation improves strong late-interaction retrievers under the same backbone, whether query-token calibration adds beyond patch grounding, which design choices matter, and how the results relate to recent systems with different training recipes. ViDoRe V2 is our primary benchmark because it reduces the extractive-query bias and saturation of the original ViDoRe benchmark; ViDoRe V1 is used as a supplementary robustness check [10, 24].

4.1 Experimental Setup

Benchmarks and metrics. Our primary evaluation uses four ViDoRe V2 subsets: ESG Reports, Biomedical Lectures, Economics Reports, and human-labeled ESG Reports. They cover non-extractive queries, long-form evidence, and cross-document information needs. We also report average results on the more extractive ViDoRe V1 benchmark. Following prior VDR work [8, 10], we report nDCG@1 and nDCG@5 as percentages.

Models. Our main comparison is same-backbone adaptation. *ColPali-v1.3* and *ColQwen2.5-v0.2* are public late-interaction checkpoints with equal-token MaxSim aggregation. *BiFocus-ColPali* and *BiFocus-ColQwen* apply our patch adaptation and query-token calibration to the corresponding checkpoints. BiCLIP, SigLIP, BiSigLIP, and DSE are included as reference baselines under the same evaluation protocol.

Training protocol. Unless otherwise stated, BiFocus follows the staged instantiation in Sec. 3. Stage I starts from a public late-interaction checkpoint and adapts the retriever with bbox-derived patch grounding. Stage II freezes the grounded retriever and trains a lightweight query-token scorer from evidence-induced token credits. Both branches are derived from the same evidence target, but staged optimization avoids unstable simultaneous training.

Evidence construction and filtering. We repurpose BBox-DocVQA boxes as cross-dataset page-internal supervision [44]; ViDoRe pages have no boxes, so retrieval results test whether this supervision transfers beyond the evidence dataset. Each valid box is clamped to the original page, directly resized to the retriever canvas, and converted into a 32×32 soft target by fractional box-patch overlap. We remove malformed boxes, missing images, pages with no valid mapped boxes, and targets with no positive mass. We further filter degenerate maps with fewer than four positive patches, overly broad maps with more than 768 positive patches, or total soft evidence mass above 700. For full training variants, P70, P60, P50, and P40 denote progressively more compact evidence subsets selected by positive-patch count, total soft mass, and positive-patch ratio over the 1024-patch grid. The main ColPali run uses P50, balancing localization quality and coverage. This filter matters because BBox-DocVQA boxes range from short answer spans to large paragraph or table regions; overly small boxes become unstable after patch discretization, whereas overly broad boxes behave like weak page-level labels rather than local evidence.

Implementation details. All retrievers use their native resolution and patch layout; for ColPali this is 448×448 with $N = 1024$ patches. Patch grounding uses soft bbox-overlap targets, cosine alignment,

$\lambda_p = 0.05$, $\tau_p = 0.05$, fp16 training, learning rate 2×10^{-5} , and the 4k-step checkpoint. The query-token scorer is a two-layer MLP with hidden dimension 512 and 0.067M parameters, trained on 1k grounded examples for 500 steps with the retriever frozen, batch size 2, learning rate 3×10^{-4} , $\tau_q = 0.2$, $\tau_t = 0.05$, and inference interpolation $\gamma = 0.2$. For the TF-IDF/lexical baseline in Fig. 3, token logits are computed from query-term TF-IDF statistics on the same grounded training-query corpus after the same tokenizer and token filtering, then normalized and interpolated with the same τ_q and γ . The same recipe can run sequentially on a single 32GB V100 with per-device batch size 1; multiple V100s are used only for wall-clock efficiency.

4.2 Main Results on ViDoRe V2

Table 1 reports the main ViDoRe V2 results. Under same-backbone adaptation, BiFocus-ColPali improves ColPali-v1.3 from 52.56 to 58.71 nDCG@1 and from 54.07 to 57.87 nDCG@5, while BiFocus-ColQwen improves ColQwen2.5-v0.2 from 56.41 to 63.53 and from 57.97 to 62.69. The strongest average result is achieved by BiFocus-ColQwen. Gains are not uniform: for ColPali, ESG remains near parity, while Economics and ESG-Human improve most. This pattern is consistent with our motivation: when queries require locating specific financial statements, multilingual terms, or human-written information needs, page-level relevance alone is often too coarse, and evidence-aware token-patch calibration becomes more useful. Overall, evidence-grounded calibration transfers across backbones and helps most on V2, where retrieval often depends on localized, non-extractive evidence. These gains require no evidence boxes at evaluation time.

4.3 Supplementary Results on ViDoRe V1

Table 2 reports average ViDoRe V1 results. Improvements are modest: BiFocus-ColPali improves ColPali-v1.3 by 0.42 nDCG@1 and 0.34 nDCG@5, while BiFocus-ColQwen improves ColQwen2.5-v0.2 by 0.17 nDCG@1 and 0.27 nDCG@5. We therefore avoid overclaiming on V1. Similar to the analysis in AGREE, this result suggests that on more explicit or extractive visual-document queries, global page-level relevance labels already provide a strong signal, so fine-grained local supervision offers limited additional gain while preserving competitive performance [8]. The main observation is therefore not that BiFocus dramatically improves V1, but that it remains stable on V1 while producing substantial improvements on V2.

4.4 Comparison with Recent Literature

Table 3 gives a contextual literature comparison with AGREE and COLMATE. We emphasize that these numbers are not a strict head-to-head comparison under identical training budgets. AGREE uses MLLM attention as proxy local supervision and trains a retriever with a global-local objective [8]. COLMATE redesigns the training pipeline with masked OCR language modeling, masked contrastive learning, and TopKSim, and reports results after a substantially different pretraining/fine-tuning recipe [26]. In contrast, our main claim is a same-backbone adaptation claim: starting from public ColPali/ColQwen checkpoints, explicit bbox evidence can calibrate both patch grounding and query-token contribution.

Table 1: Main results on ViDoRe V2. Primary comparisons are same-backbone public-checkpoint adaptations from ColPali-v1.3 to BiFocus-ColPali and from ColQwen2.5-v0.2 to BiFocus-ColQwen. The Avg. columns are separated by a thicker rule to highlight overall performance. The best score in each metric column is bolded, and the second-best score is underlined. Light-blue rows denote our models.

Model	ESG		Biomedical		Economics		ESG-Human		Avg.	
	@1	@5	@1	@5	@1	@5	@1	@5	@1	@5
<i>Vision-Language Models</i>										
BiCLIP	10.53	11.38	20.16	22.39	26.29	22.83	17.31	22.74	18.57	19.84
SigLIP (0-shot)	26.75	28.19	20.16	24.30	18.97	16.34	19.23	23.07	21.28	22.98
BiSigLIP	29.83	34.92	25.78	29.05	30.60	28.23	34.62	45.83	30.21	34.51
<i>Visual Retrievers</i>										
DSE	47.37	<u>55.15</u>	52.66	55.29	62.50	52.80	57.69	61.74	55.06	56.25
ColPali-v1.3	<u>55.70</u>	55.06	51.72	55.62	52.16	48.51	50.64	57.10	52.56	54.07
ColQwen2.5-v0.2	50.88	53.03	<u>56.25</u>	<u>59.16</u>	59.91	53.96	58.61	<u>65.76</u>	56.41	<u>57.97</u>
<i>Ours: Evidence-Grounded Adaptation</i>										
BiFocus-ColPali	55.33	54.90	53.78	58.16	<u>65.16</u>	<u>55.12</u>	<u>60.56</u>	63.29	<u>58.71</u> ↑6.15	57.87 ↑3.80
BiFocus-ColQwen	56.93	56.77	61.52	63.79	67.46	60.36	68.21	69.82	63.53 ↑7.12	62.69 ↑4.72

Table 2: Average results on ViDoRe V1. V1 is used as a robustness check rather than the main source of performance claims.

Model	nDCG@1	nDCG@5
ColPali-v1.3	77.17	83.90
BiFocus-ColPali	77.59 ↑0.42	84.24 ↑0.34
ColQwen2.5-v0.2	83.56	89.45
BiFocus-ColQwen	83.73 ↑0.17	89.72 ↑0.27

Table 3: Contextual literature comparison with recent VDR methods on ViDoRe V2. AGREE and COLMATE are included to situate BiFocus within recent VDR literature rather than as strictly fair same-budget baselines. “–” indicates that the cited paper reports nDCG@5 only for that entry.

Model	Main supervision / recipe	Avg.@1	Avg.@5
AGREEPali [8]	MLLM attention	59.12	57.75
AGREEQwen2.5 [8]	MLLM attention	61.84	61.54
COLMATE-Pali [26]	MOLM+MaskedCL+TopKSim	–	57.61
COLMATE-Qwen [26]	MOLM+MaskedCL+TopKSim	–	60.60
BiFocus-ColPali	explicit boxes + token credits	58.71	57.87
BiFocus-ColQwen	explicit boxes + token credits	63.53	62.69

This contextual comparison is still informative. AGREE supports the broader conclusion that fine-grained local guidance improves VDR, but its local labels are teacher-attention proxies and mainly supervise the page side. COLMATE shows that document-specific representation learning and training-time similarity design can

Table 4: Component ablation on ColPali-v1.3. Patch grounding gives the main improvement on ViDoRe V2; query-token weighting further calibrates the grounded retriever.

Setting	ViDoRe V2			ViDoRe V1	
	@1	@5	Δ@1	@1	@5
ColPali-v1.3	52.56	54.07	–	77.17	83.90
+ BBox patch grounding	57.19	56.23	↑4.63	77.06	83.50
+ Query-token weighting	58.71	57.87	↑6.15	77.59	84.24

improve retrieval, but it is closer to a new training recipe than to lightweight evidence calibration. BiFocus addresses a different and complementary problem: how to inject explicit page-internal evidence into already strong public late-interaction retrievers and use it to calibrate both where to look and which query tokens should drive the score.

4.5 Component Ablation

Table 4 decomposes BiFocus on ColPali-v1.3. On ViDoRe V2, bbox patch grounding raises nDCG@1 from 52.56 to 57.19 and nDCG@5 from 54.07 to 56.23. Query-token weighting further reaches 58.71 nDCG@1 and 57.87 nDCG@5. On ViDoRe V1, patch grounding alone is slightly below the baseline, but query-token weighting recovers and improves performance. This behavior matches our hypothesis: patch grounding provides the main gain under complex V2 queries, while token weighting is a calibration layer that refines the grounded retriever and helps prevent over-specialization to local evidence.

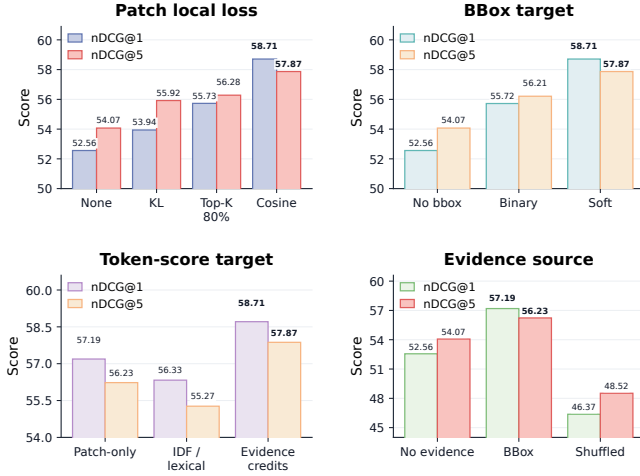


Figure 3: Design ablations on ViDoRe V2. Cosine patch alignment, soft overlap targets, evidence-induced token credits, and correct bbox evidence are all important for reliable calibration.

4.6 Design Ablations: What Matters?

Figure 3 summarizes four design choices. First, cosine alignment performs best among local patch losses. This ablation follows the local-alignment analysis in AGREE, where cosine directional alignment is more robust than strict distribution matching because it emphasizes salient-region agreement without forcing every low-salience background patch to match [8]. Under explicit-box supervision, we observe the same trend: KL improves nDCG@5 but is weaker on nDCG@1, Top-k remains threshold-dependent, and cosine gives the strongest result. Second, soft overlap targets outperform binary patch labels, confirming that fractional box–patch overlap better handles evidence boundaries. Third, evidence-induced token credits outperform the TF-IDF/lexical heuristic. TF-IDF measures corpus-level lexical rarity, whereas evidence credits measure whether a token actually reaches the visual evidence in a particular query–page match; this distinction matters for numbers, abbreviations, table headers, and relation cues that may be visually decisive without being globally rare. Finally, shuffled boxes severely hurt performance, showing that improvements come from correct evidence localization rather than from additional fine-tuning alone.

4.7 Hyperparameters and Evidence Quality

Figure 4 studies λ_p , γ , and evidence-quality filtering. The best patch-grounding weight is $\lambda_p = 0.05$: smaller values underuse the evidence signal, while larger values distort the global retrieval space. The best token-weighting strength is $\gamma = 0.2$, balancing learned weights with the stability of the original MaxSim sum. For evidence filtering, P50 performs best; weaker filtering leaves broad or mixed boxes, whereas overly aggressive filtering reduces training coverage. These trends support a calibration view of BiFocus: local evidence should regularize the late-interaction space, but it should not overwhelm the original retrieval objective or collapse supervision onto a small set of easy boxes.

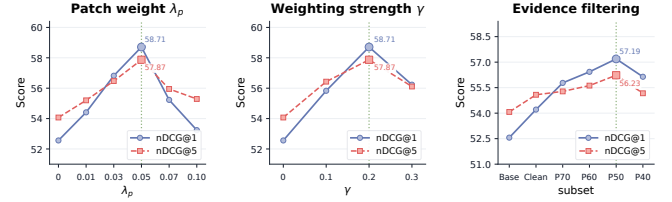


Figure 4: Hyperparameter and evidence-quality sensitivity on ViDoRe V2. The best settings are $\lambda_p = 0.05$, $\gamma = 0.2$, and the P50 evidence subset.

Table 5: Aggregate case-mining statistics on ViDoRe V2. “Improved” means the ground-truth rank becomes better; “Top-1 gain” promotes a ground-truth page to rank 1; “Top-1 fix” corrects a non-relevant ColPali top-1 page.

Subset	Q.	Impr.	T1-Gain	T1-Fix	Avg. GT Rank
ESG	228	60	20	23	14.07→11.78
Biomedical	640	163	53	51	15.06→11.69
Economics	232	78	44	48	3.64→2.69
ESG-Human	52	17	7	5	10.31→6.63
Total / Avg.	1152	318	124	127	12.35→9.67

4.8 Qualitative Analysis and Case Study

Figure 5 visualizes the two calibration signals on a BBox-DocVQA example. The annotated box is projected to a soft patch target over the relevant table region. Compared with ColPali, BiFocus concentrates the patch relevance map around the boxed evidence and improves patch-target cosine from 0.110 to 0.209. The induced token credits also assign larger mass to evidence-bearing tokens such as “DS” and “AI”, as well as the ordering cue “then”; the learned scorer mirrors this pattern with a token credit–weight correlation of 0.87. This suggests that the scorer learns evidence-grounded token weighting rather than generic lexical salience alone.

Figure 6 gives a ViDoRe V2 retrieval case. This benchmark example has no evidence boxes at evaluation time and is used only to inspect ranking behavior. For the French query about the implications of a stronger U.S. dollar for PFRD debt, ColPali retrieves a non-relevant page at rank 1, while BiFocus retrieves a relevant page and improves the ground-truth rank from 77 to 1. The learned weights emphasize finance-related terms such as “dollar”, “dette”, and “PFRD”, illustrating how token calibration can change retrieval outcomes on a realistic multilingual query.

We also mine cases over all four ViDoRe V2 subsets. For each query, we compare the ground-truth ranks of ColPali-v1.3 and our ColPali variant, and count improved ranks, top-1 gains, and top-1 fixes. Table 5 shows that BiFocus improves 318 out of 1152 queries, promotes 124 queries to rank 1, fixes 127 wrong top-1 cases, and improves the average ground-truth rank from 12.35 to 9.67. Thus, the retrieval case in Figure 6 is not isolated.

Together, the visualizations separate mechanism from retrieval behavior. BBox-DocVQA examples diagnose whether internal maps move toward annotated support regions, whereas ViDoRe V2 cases

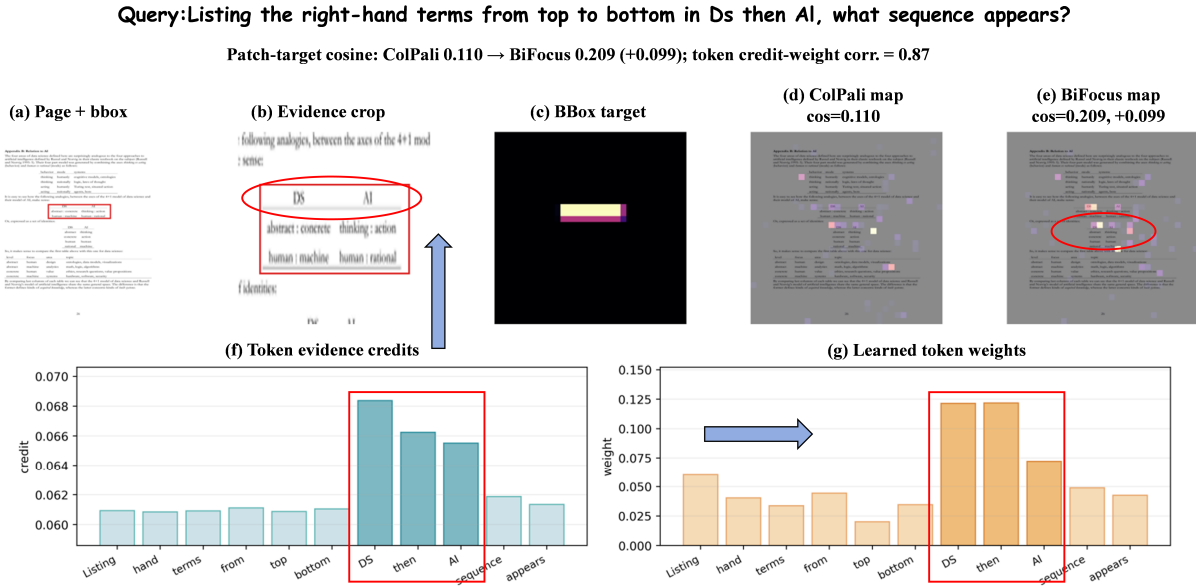


Figure 5: Mechanism visualization on a BBox-DocVQA example. The query asks for the sequence of terms under the “DS” and “AI” columns. BiFocus shifts the patch relevance map toward the annotated evidence region and assigns larger weights to evidence-bearing query tokens. The patch-target cosine improves from 0.110 to 0.209, and the learned token weights correlate strongly with evidence-induced token credits.

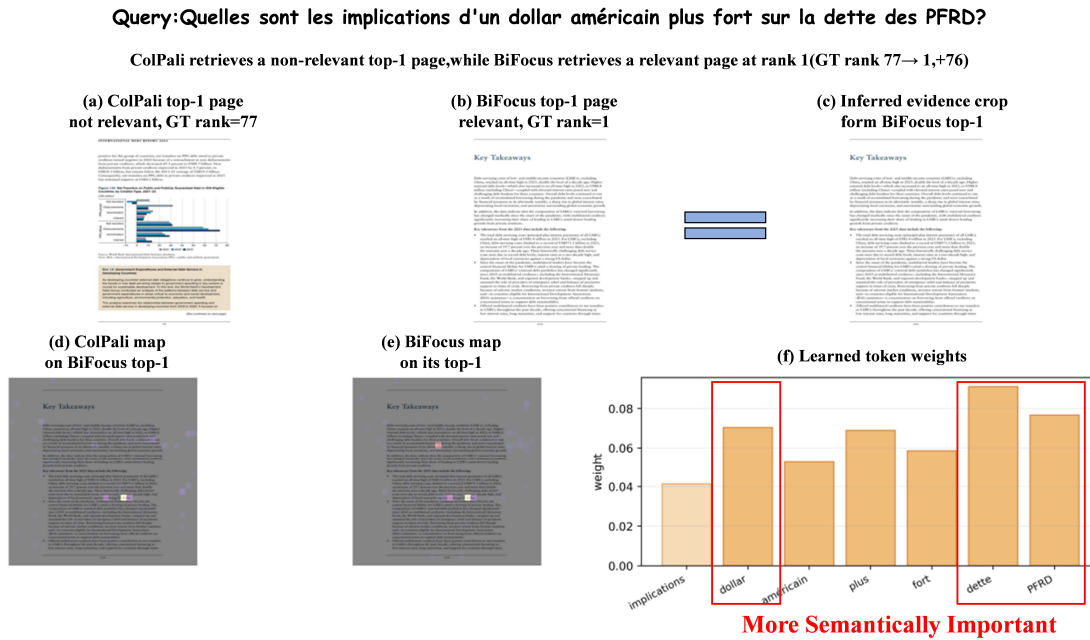


Figure 6: Retrieval case study on ViDoRe V2. For the same multilingual economics query, ColPali retrieves a non-relevant page at rank 1, whereas BiFocus retrieves a relevant page at rank 1, improving the ground-truth rank from 77 to 1. The learned token weights emphasize finance-related query terms such as “dollar”, “dette”, and “PFRD”.

test whether the induced calibration generalizes to unseen corpora and queries without test-time boxes.

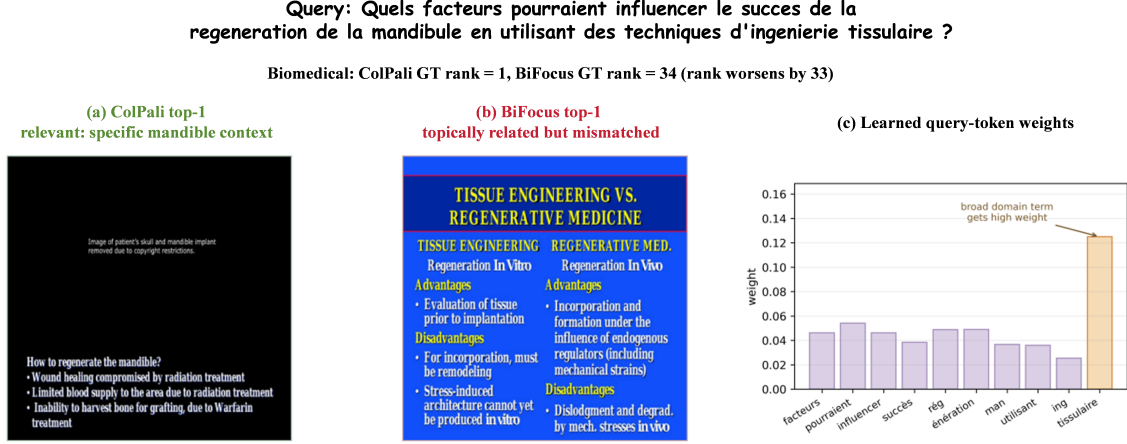


Figure 7: Informative failure case on ViDoRe V2. ColPali retrieves the ground-truth page at rank 1, while BiFocus ranks it at 34. The BiFocus top-1 page is topically related to tissue engineering but mismatched with the specific mandible-regeneration context. The learned token weights assign a high weight to the broad domain term “tissulaire”, illustrating that evidence-induced token credits can emphasize broad domain evidence when relevance depends on finer-grained compositional semantics.

Table 6: Efficiency comparison between ColPali-v1.3 and BiFocus-ColPali. Latency is measured on one NVIDIA V100 using sampled ViDoRe V2 queries and pages.

Model	Query Enc. ms/query	Page Enc. ms/page	Scoring ms/(q,page)	Extra Params
ColPali-v1.3	21.26	718.25	0.0017	–
BiFocus-ColPali	23.43	768.26	0.0018	0.067M

4.9 Efficiency Analysis

A practical advantage of BiFocus is that evidence supervision is only used for training and calibration. At inference time, the retriever still takes a textual query and page screenshots, without boxes, OCR regions, MLLM teacher attention, or gallery-side refinement. We measure latency on 64 queries and 128 pages from ViDoRe V2 using one NVIDIA V100 GPU; results are in Table 6.

BiFocus adds modest overhead: 1.10× query-encoding latency, 1.07× page-encoding latency, and 1.07× scoring latency. The scorer has only 0.067M parameters and costs 0.0067 ms per query. The page encoder and embedding shape are unchanged, so page-index storage does not increase; both models require approximately 257.8 KB per page in fp16. Thus, the accuracy gains come from a light-weight query-side calibration step while preserving the page-level retrieval interface.

4.10 Failure Case and Limitations

Although BiFocus improves overall accuracy, evidence-induced token calibration is a calibration signal rather than a replacement for semantic query understanding. Figure 7 shows a Biomedical case where ColPali ranks the ground-truth mandible-regeneration page first, but BiFocus ranks a topically related tissue-engineering slide first and the ground truth at 34. The scorer gives high weight to the broad term “tissulaire”, which is relevant to the domain but insufficient to identify the specific context. Thus, local evidence

coverage can diverge from compositional query semantics when relevance depends on finer or distributed cues. Future work can combine evidence-induced credits with semantic token importance, multi-region aggregation, or robustness to noisy evidence.

4.11 Discussion

The experiments yield three takeaways. Explicit boxes provide effective local supervision when page-level labels are too coarse; query-token weighting complements patch grounding by calibrating which tokens dominate MaxSim aggregation; and evidence quality matters because broad or shuffled boxes can hurt retrieval. Accordingly, we use BBox-DocVQA visualizations as evidence-alignment diagnostics and ViDoRe V2 examples as retrieval case studies, not localization claims on datasets without human evidence boxes.

5 Conclusion

We introduced BiFocus, an evidence-grounded calibration framework for screenshot-based late-interaction VDR. It addresses a supervision asymmetry in ColPali-style retrievers by using BBox-DocVQA evidence boxes to calibrate both page-side grounding and query-token contribution. At inference, BiFocus requires no boxes, OCR regions, MLLM attention maps, or external teachers.

BiFocus delivers substantial same-backbone gains on ViDoRe V2 for ColPali-v1.3 and ColQwen2.5-v0.2 while preserving strong ViDoRe V1 performance. Ablations show that patch grounding provides the main gain, token weighting further calibrates the retriever, and evidence quality matters. These results support calibrating both visual evidence and textual contribution in late-interaction VDR.

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GenAI Usage Disclosure

Generative AI (GenAI) tools were utilized in the preparation of this manuscript to assist with coding and language polishing. The core research ideas, experimental design, analysis, and the overall drafting and writing of the paper were conducted by the authors without the involvement of GenAI.

References

- [1] Srikar Appalaraju, Bhavan Jasani, Bhargava Urala Kota, Yusheng Xie, and R. Manmatha. 2021. DocFormer: End-to-End Transformer for Document Understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 993–1003.
- [2] S. Archish, Ankit Garg, Kirankumar Shiragur, and Neeraj Kayal. 2025. Incorporating Token Importance in Multi-Vector Retrieval. *arXiv preprint arXiv:2511.16106* (2025). <https://arxiv.org/abs/2511.16106>
- [3] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, et al. 2025. Qwen2.5-VL Technical Report. *arXiv preprint arXiv:2502.13923* (2025). <https://arxiv.org/abs/2502.13923>
- [4] Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, Mir Rosenberg, Xia Song, Alina Stoica, Saurabh Tiwary, and Tong Wang. 2016. MS MARCO: A Human Generated Machine Reading Comprehension Dataset. In *Advances in Neural Information Processing Systems*. <https://arxiv.org/abs/1611.09268>
- [5] Lucas Beyer, Andreas Steiner, André Susano Pinto, Alexander Kolesnikov, Xiao Wang, Daniel Salz, Maxim Neumann, Ibrahim Alabdulmohsin, Michael Tschanen, Emanuele Bugliarello, et al. 2024. PaliGemma: A Versatile 3B VLM for Transfer. *arXiv preprint arXiv:2407.07726* (2024). <https://arxiv.org/abs/2407.07726>
- [6] Lukas Blecher, Guillem Cucurull, Thomas Scialom, and Robert Stojnic. 2024. Nougat: Neural Optical Understanding for Academic Documents. In *Proceedings of the Twelfth International Conference on Learning Representations*. <https://openreview.net/forum?id=Q1y5VhVGB>
- [7] Xi Chen, Xiao Wang, Lucas Beyer, Alexander Kolesnikov, Jialin Wu, Paul Voigtlaender, Basil Mustafa, Sebastian Goodman, Ibrahim Alabdulmohsin, Piotr Padlewski, et al. 2022. PaLI: A Jointly-Scaled Multilingual Language-Image Model. *arXiv preprint arXiv:2209.06794* (2022). <https://arxiv.org/abs/2209.06794>
- [8] Wanqing Cui, Wei Huang, Yazhi Guo, Yibo Hu, Meiguang Jin, Junfeng Ma, and Keping Bi. 2025. Attention Grounded Enhancement for Visual Document Retrieval. *arXiv preprint arXiv:2511.13415* (2025). <https://arxiv.org/abs/2511.13415>
- [9] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 4171–4186. doi:10.18653/v1/N19-1423
- [10] Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. 2025. ColPali: Efficient Document Retrieval with Vision Language Models. In *Proceedings of the Thirteenth International Conference on Learning Representations*. <https://openreview.net/forum?id=ogjBpZ8uSi>
- [11] Thibault Formal, Benjamin Piwowarski, and Stéphane Clinchant. 2021. SPLADE: Sparse Lexical and Expansion Model for First Stage Ranking. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2288–2292. doi:10.1145/3404835.3463098
- [12] Luyu Gao, Zhuyun Dai, and Jamie Callan. 2021. COIL: Revisit Exact Lexical Match in Information Retrieval with Contextualized Inverted List. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. 3030–3042. doi:10.18653/v1/2021.naacl-main.241
- [13] Agathoklis Georgiou. 2025. Spatially-Grounded Document Retrieval via Patch-to-Region Relevance Propagation. *arXiv preprint arXiv:2512.02660* (2025). <https://arxiv.org/abs/2512.02660>
- [14] Yupan Huang, Tengchao Lv, Lei Cui, Yutong Lu, and Furu Wei. 2022. Lay-outLMv3: Pre-training for Document AI with Unified Text and Image Masking. In *Proceedings of the 30th ACM International Conference on Multimedia*. 4083–4091. doi:10.1145/3503161.3548112
- [15] Chao Jia, Yinfei Yang, Ye Xia, Yuning Chen, Zarana Parekh, Hieu Pham, Quoc V. Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. 2021. Scaling Up Visual and Vision-Language Representation Learning with Noisy Text Supervision. In *Proceedings of the 38th International Conference on Machine Learning*. 4904–4916.
- [16] Hyukkyu Kang, Injung Kim, and Wook-Shin Han. 2025. TRIAL: Token Relations and Importance Aware Late-interaction for Accurate Text Retrieval. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 16864–16877. doi:10.18653/v1/2025.emnlp-main.854
- [17] Omar Khattab and Matei Zaharia. 2020. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. 39–48. doi:10.1145/3397271.3401075
- [18] Geewook Kim, Teakgyu Hong, Moonbin Yim, Jeongyeon Nam, Jinyoung Park, Jinyeong Yim, Wonseok Hwang, Sangdo Yun, Dongyoon Han, and Seunghyun Park. 2022. OCR-free Document Understanding Transformer. In *Proceedings of the European Conference on Computer Vision*. 498–517. doi:10.1007/978-3-031-19815-1_29
- [19] Carlos Lassance, Hervé Déjean, Thibault Formal, and Stéphane Clinchant. 2024. SPLADE-v3: New Baselines for SPLADE. *arXiv preprint arXiv:2403.06789* (2024). <https://arxiv.org/abs/2403.06789>
- [20] Kenton Lee, Mandar Joshi, Julia Turc, Hexiang Hu, Fangyu Liu, Julian Eisenschlos, Urvashi Khandelwal, Peter Shaw, Ming-Wei Chang, and Kristina Toutanova. 2023. Pix2Struct: Screenshot Parsing as Pretraining for Visual Language Understanding. In *Proceedings of the 40th International Conference on Machine Learning*. 18893–18912.
- [21] Yinglu Li, Zhiying Lu, Zhihang Liu, Yiwei Sun, Chuanbin Liu, and Hongtao Xie. 2025. RegionRAG: Region-level Retrieval-Augmented Generation for Visual Document Understanding. *arXiv preprint arXiv:2510.27261* (2025). <https://arxiv.org/abs/2510.27261>
- [22] Xueguang Ma, Sheng-Chieh Lin, Minghan Li, Wenhu Chen, and Jimmy Lin. 2024. Unifying Multimodal Retrieval via Document Screenshot Embedding. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 6492–6505. doi:10.18653/v1/2024.emnlp-main.370
- [23] Sean MacAvaney, Andrew Yates, Arman Cohan, and Nazli Goharian. 2019. CEDR: Contextualized Embeddings for Document Ranking. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. 1101–1104. doi:10.1145/3331184.3331317
- [24] Quentin Macé, António Loison, and Manuel Faysse. 2025. ViDoRe Benchmark V2: Raising the Bar for Visual Retrieval. *arXiv preprint arXiv:2505.17166* (2025). <https://arxiv.org/abs/2505.17166>
- [25] Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. 2022. ChartQA: A Benchmark for Question Answering about Charts with Visual and Logical Reasoning. In *Findings of the Association for Computational Linguistics: ACL 2022*. 2263–2279. doi:10.18653/v1/2022.findings-acl.177
- [26] Ahmed Masry, Megh Thakkar, Patrice Bechard, Satwik Tejaswi Madhusudhan, Rabiul Awal, Shambhavi Mishra, Akshay Kalkunte Suresh, Srivatsava Daruru, Enamul Hoque, Spandana Gella, Torsten Scholak, and Sai Rajeswar. 2025. COLMATE: Contrastive Late Interaction and Masked Text for Multimodal Document Retrieval. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*. 2071–2080. <https://aclanthology.org/2025.emnlp-industry.145/>
- [27] Minesh Mathew, Viraj Bagal, Rubén Tito, Dimosthenis Karatzas, Ernest Valveny, and C. V. Jawahar. 2022. InfographicVQA. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2582–2591.
- [28] Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. 2021. DocVQA: A Dataset for VQA on Document Images. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 2200–2209.
- [29] Anand Mishra, Shashank Shekhar, Ajeet Kumar Singh, and Anirban Chakraborty. 2019. OCR-VQA: Visual Question Answering by Reading Text in Images. In *Proceedings of the 2019 International Conference on Document Analysis and Recognition*. 947–952. doi:10.1109/ICDAR.2019.00156
- [30] Rodrigo Nogueira, Zhiying Jiang, and Jimmy Lin. 2020. Document Ranking with a Pretrained Sequence-to-Sequence Model. *arXiv preprint arXiv:2003.06713* (2020). <https://arxiv.org/abs/2003.06713>
- [31] Raj Nath Patel and Sourav Dutta. 2026. ColBERT-Att: Late-Interaction Meets Attention for Enhanced Retrieval. *arXiv preprint arXiv:2603.25248* (2026). <https://arxiv.org/abs/2603.25248>
- [32] Dipannita Podder, Jialu H. Paik, and Pabitra Mitra. 2024. Learning Query Token Importance for Effective Document Retrieval with Verbose Queries. In *Proceedings of the 16th Annual Meeting of the Forum for Information Retrieval Evaluation*. 67–75. doi:10.1145/3734947.3734954
- [33] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning*. 8748–8763.
- [34] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*. 3982–3992. doi:10.18653/v1/D19-1410
- [35] Stephen E. Robertson and Steve Walker. 1994. Some Simple Effective Approximations to the 2-Poisson Model for Probabilistic Weighted Retrieval. In *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. 232–241.
- [36] Keshav Santhanam, Omar Khattab, Jon Saad-Falcon, Christopher Potts, and Matei Zaharia. 2022. ColBERTv2: Efficient and Effective Retrieval via Lightweight Late Interaction. In *Proceedings of the 2022 Conference of the North American Chapter*

- of the Association for Computational Linguistics: Human Language Technologies. 3715–3734. doi:10.18653/v1/2022.naacl-main.272
- [37] Ryota Tanaka, Taichi Iki, Taku Hasegawa, Kyosuke Nishida, Kuniko Saito, and Jun Suzuki. 2025. VDocRAG: Retrieval-Augmented Generation over Visually-Rich Documents. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 24827–24837. doi:10.1109/CVPR52734.2025.02312
- [38] Ryota Tanaka, Kyosuke Nishida, Taro Yoshida, Koki Ohyama, Hiroaki Oiwa, and Hiroyuki Shindo. 2023. SlideVQA: A Dataset for Document Visual Question Answering on Multiple Images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 13643–13651. doi:10.1609/aaai.v37i11.26564
- [39] Nandan Thakur, Nils Reimers, Andreas Rueckle, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. In *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*. <https://arxiv.org/abs/2104.08663>
- [40] Rubèn Tito, Dimosthenis Karatzas, and Ernest Valveny. 2023. Hierarchical Multimodal Transformers for Multi-Page DocVQA. *Pattern Recognition* 144 (2023), 109834. doi:10.1016/j.patcog.2023.109834
- [41] Yiheng Xu, Minghao Li, Lei Cui, Shaohan Huang, Furu Wei, and Ming Zhou. 2020. LayoutLM: Pre-training of Text and Layout for Document Image Understanding. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 1192–1200. doi:10.1145/3394486.3403172
- [42] Yang Xu, Yiheng Xu, Tengchao Lv, Lei Cui, Furu Wei Zhang, Guoxin Wang, Yijuan Lu, Dinei Florencio, Cha Zhang, Wanxiang Che, Min Zhang, and Lidong Zhou. 2021. LayoutLMv2: Multi-modal Pre-training for Visually-Rich Document Understanding. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing*. 2579–2591. doi:10.18653/v1/2021.acl-long.201
- [43] Hao Yang, Yifan Ji, Zhipeng Xu, Zhenghao Liu, Yukun Yan, Zulong Chen, Shuo Wang, Yu Gu, and Ge Yu. 2026. ReAlign: Optimizing the Visual Document Retriever with Reasoning-Guided Fine-Grained Alignment. *arXiv preprint arXiv:2604.07419* (2026). <https://arxiv.org/abs/2604.07419>
- [44] Wenhan Yu, Wang Chen, Guanqiang Qi, Weikang Li, Yang Li, Lei Sha, Deguo Xia, and Jizhou Huang. 2025. BBox DocVQA: A Large Scale Bounding Box Grounded Dataset for Enhancing Reasoning in Document Visual Question Answer. *arXiv preprint arXiv:2511.15090* (2025). <https://arxiv.org/abs/2511.15090>
- [45] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. Sigmoid Loss for Language Image Pre-Training. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 11975–11986.
- [46] Yuanyang Zhang, Yijie Lin, Xinhang Wan, Jie Xu, Li Yao, Weiqing Yan, and Chang Tang. 2026. Structure-Aware Conditional Diffusion Generation for Incomplete Multi-View Clustering. *IEEE Transactions on Knowledge and Data Engineering* 38, 5 (2026), 3168–3181.
- [47] Yuanyang Zhang, Yijie Lin, Weiqing Yan, Li Yao, Xinhang Wan, Guangyuan Li, Chao Zhang, Guanzhou Ke, and Jie Xu. 2025. Incomplete Multi-view Clustering via Diffusion Contrastive Generation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 22650–22658.
- [48] Yuanyang Zhang, Xinhang Wan, Chao Zhang, Jie Xu, Cunjian Chen, Tien-Tsin Wong, Li Yao, and Yijie Lin. 2026. Mamba-driven multi-view discriminative clustering via global-local cross-view sequence modeling. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 40. 28519–28527.
- [49] Yuanyang Zhang, Weiqing Yan, Chang Tang, Wujie Zhou, and Jian Jin. 2025. Multi-branch Space Sharing Feature Aggregation for contrastive multi-view clustering. *Pattern Recognition* 167 (2025), 111704.
- [50] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Yuchen Duan, Hao Tian, Weijie Su, Jie Shao, et al. 2025. InternVL3: Exploring Advanced Training and Test-Time Recipes for Open-Source Multimodal Models. *arXiv preprint arXiv:2504.10479* (2025). <https://arxiv.org/abs/2504.10479>